

Amazon Review Integrity and Behavioral Analysis Using Big Data Methods

Kireeti Josyula | ADSP 31013 | 12/12/2025

Executive Summary

- Large-scale analysis of Amazon review and product metadata (2012–2024) using Spark on GCS
- Explored review growth, product engagement, pricing effects, reviewer behavior, and text uniqueness
- Review volume exhibits strong temporal growth with clear seasonality and post-COVID peaks (2019–2021)
- High review counts are driven by a small number of highly engaged products and generalist reviewers
- Near-duplicate detection (MinHash + LSH) reveals frequent title reuse but largely original review texts
- Similarity-based detection is significantly less effective on recent reviews, suggesting increased linguistic diversity and potential AI-driven paraphrasing in newer content.

Data Cleanup and Initial Analysis

- Reviews Data Cleaning

- Filter out purchases that are not verified (i.e `verified_purchase == false`)
- Convert time from milliseconds into standard date formats.
- Filter out reviews that do not have a valid rating (i.e., null values or ratings outside `[1,5]`).
- Removing rows with missing/empty review text and missing review dates
- Drop exact duplicate reviews

- Meta Data Cleaning

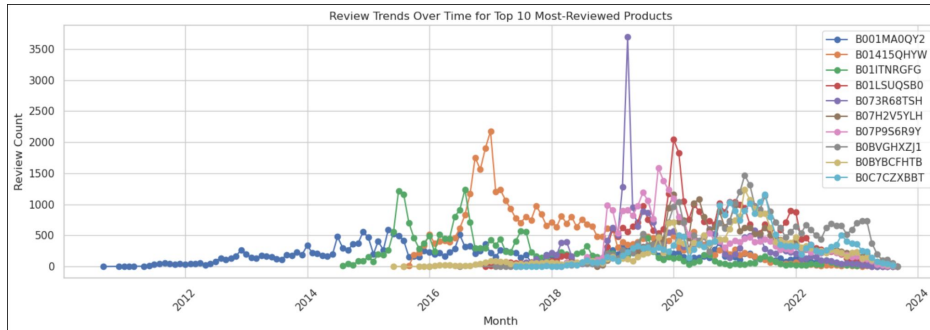
- Drop unnecessary columns for analysis (author, bought_together, subtitle, description, store)
- Convert price string into numeric price
- Filter rows where price, rating number or average number are invalid
- Deduplicate records across title, main category and prices.

Growth of Reviews by Time



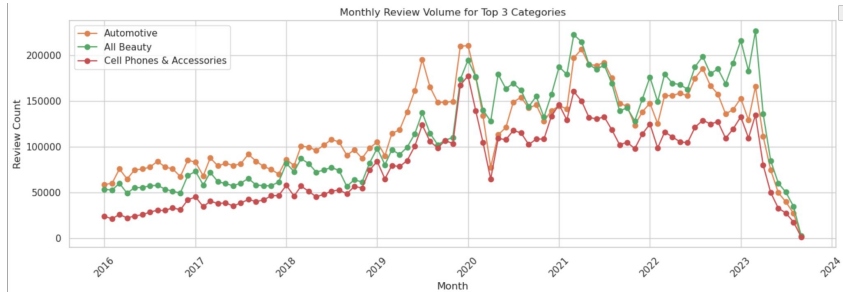
- Timeline of the data: 2012 to 2024
- Significant peaks between 2019 and 2021,
 - Peak in 2021 might reflect post COVID-19 surge in Amazon reliance
 - Peak in 2019 likely reflects Amazon's gradual platform adoption
- Early years show low volume (~20,000 reviews per month) → Incomplete early data coverage
- Late years show low volume → Incomplete ingestion of data at time of analysis
- NOTE: Timeline cuts off data from 2000 - 2012 for clarity of chart due to low review volume

Review Trends for Top Products



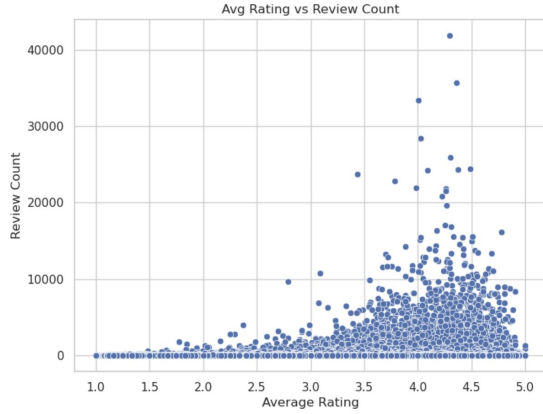
- Top 10 most reviewed products (shown in figure) exhibit highly non-uniform lifecycle patterns.
- In terms of review counts, all of the products seem to display:
 - 1) Ramp-Up phase,
 - 2) one or more sharp spikes,
 - 3) long stabilization and decline period (roughly post 2021)
- **Major Conclusion: High total review volume over analysis period does not imply consistent engagement. Many top products peak briefly and then decline**

Review Volume Trends for Top 3 Categories

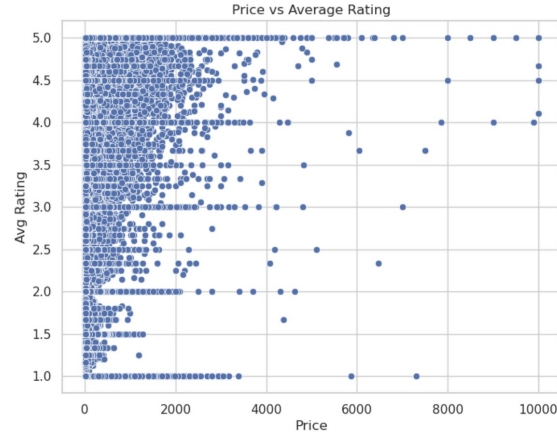


- Top 3 Review Categories:
 - Automotive (12,430,003),
 - All Beauty (11,785,871),
 - Cell Phones & Accessories (8,054,059)
- These three categories together account for a large share of all reviews in the dataset.
- All top 3 categories show late-year period spikes, indicating increased gift purchasing for holidays (Christmas, Black Friday, Prime Day)
- Conclusions: **Temporal alignment in top 3 categories suggests seasonality effects**

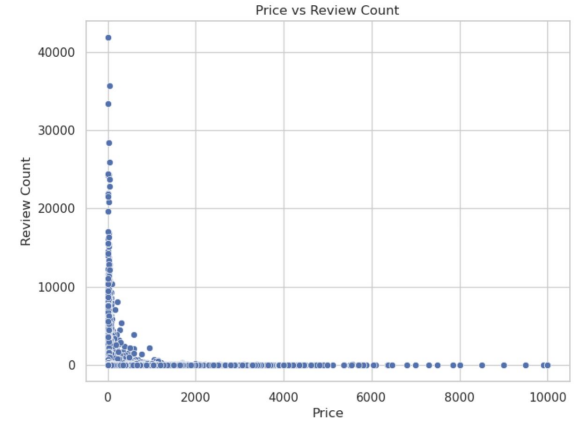
Relationships between Ratings, Prices and Reviews



Conclusion: Higher-rated products tend to receive higher review volumes.

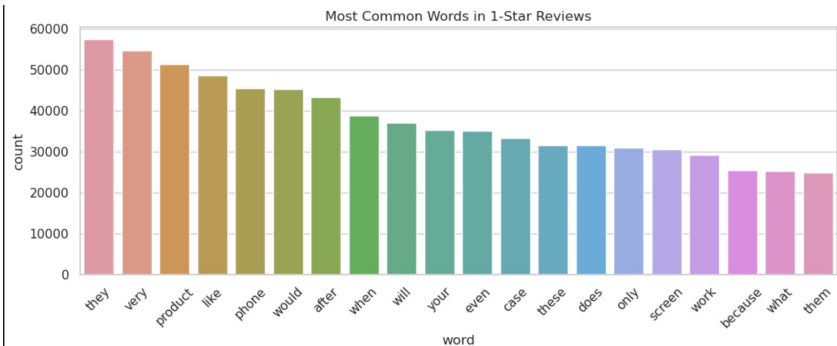
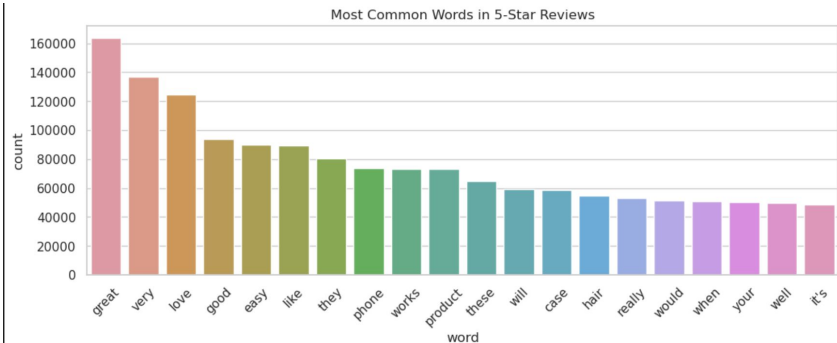


Conclusion: No relationship between product price and average rating



Conclusion: Lower-priced products receive substantially more reviews than high-priced items

Common words in 5-star vs 1-star Reviews



- Top chart: 5-Star Reviews
 - Positive reviews emphasize emotional and experiential words (“great”, “love”, “easy”).
 - Language is focused on satisfaction and usability, not problems.
- Bottom chart: 1-Star Reviews
 - Negative reviews focus on process and failure (“after”, “would”, “when”, “does”).
 - Language is more situational and explanatory, suggesting issue narratives.
- Conclusions:
 - **5-Star: Reinforcement of the “goodness” of product**
 - **1-Star: Explanation of “badness” of product**

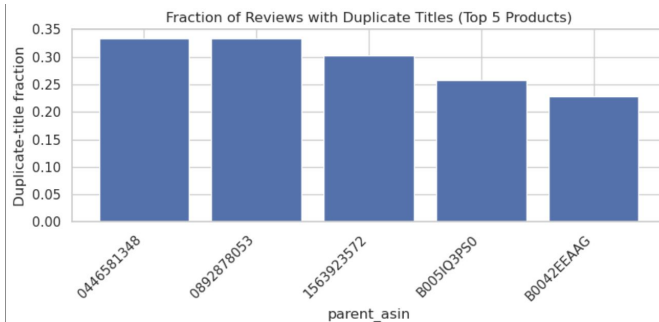
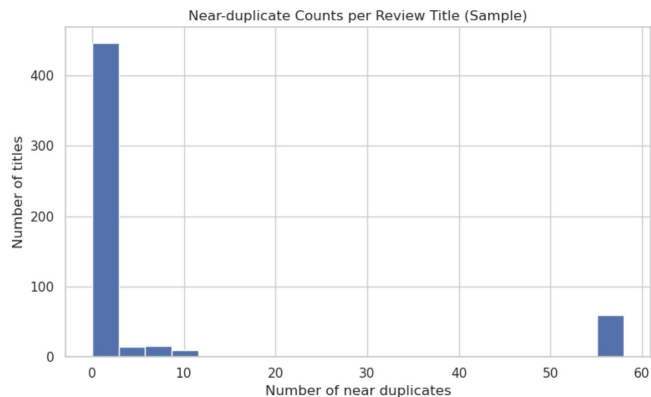
Most Active Reviewers

Table 4: Most Active Reviewers and Category Diversity (Top 20)

Rank	Reviewer (user_id)	# Reviews	# Categories
1	AFTZWAK3ZHAPCNSOT5GCKQDECBTQ	883	17
2	AGUTZC4GHLTGYHA3KBEDRF6MHB6A	827	14
3	AHH67GCRGS45Y670Z7TS7TNPVELQ	522	10
4	AEYKTZXAWOPJG5MGMKBLRJR6Q3A_1	433	14
5	AEYKTZXAWOPJG5MGMKBLRJR6Q3A_2	412	10
6	AGCP6600WFSGHQGGQF5SLTY6HRNQ	404	7
7	AGSXMEEFKFM2HDFZ4UNE67606D5LQ	390	8
8	AGFN3252BBTYUJUUDQMAGYNUT5A_1	381	13
9	AGFN3252BBTYUJUUDQMAGYNUT5A_2_1	379	11
10	AHP6Z5H7TMGB05502PIB6HVR5VRQ	377	9
11	AFZUK3MTBIBEDQOPAK30ATUOUKLA	352	14
12	AGPC7CWRQPMNQFXWLYJYQYF7DCQA	351	10
13	AGRHKDNSRJ3CT5ST75KGSCD4WA5A	353	12
14	AFP2Z2JLDXU6PYRZFQ2A4ZYPLM4Q	342	11
15	AFEYKY4PX6MMT64Q4L57K5VWZ6LA	338	8
16	AFNMU7RPT24NMJGKD72QFFYB35LQ	336	8
17	AH20SSSLGXNEFZBCUYANAUFUNBA	316	11
18	AE2K6NGRXS6NMFJFWUMEYT7N4NPA	314	11
19	AG2RNXPP7TLAF6RR7BMHMIYPGNX3Q	314	9
20	AEFTRCMDX2VGCKEW46X2A35CXKCQ	299	10

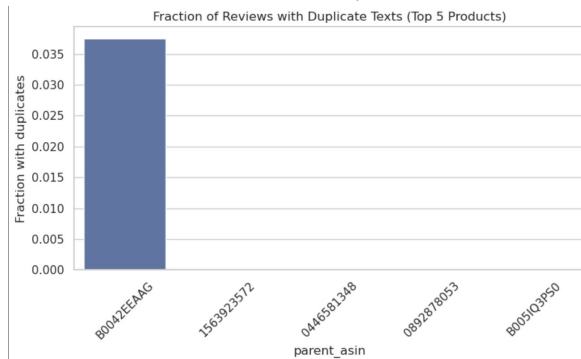
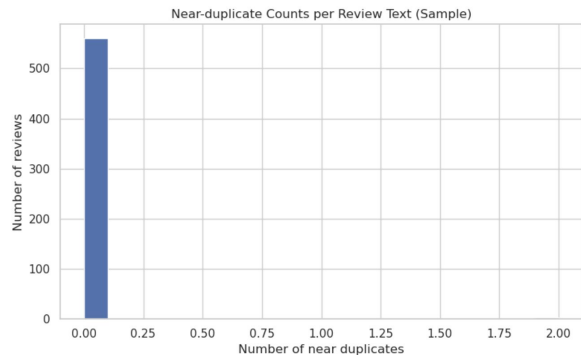
- Top reviewer by total review volume was “AFTZWAK3Z.....” With 883 reviews, followed by “AGUTZC4GHLT.....” With 827 reviews.
- Many top reviewers span ~10–17 categories
- This suggests top reviewers are often generalist reviewers, not confined to just 1–2 categories

Near-Duplicate Review Titles (Books Category)



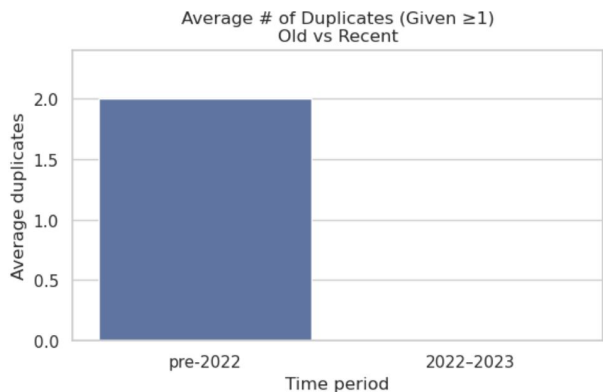
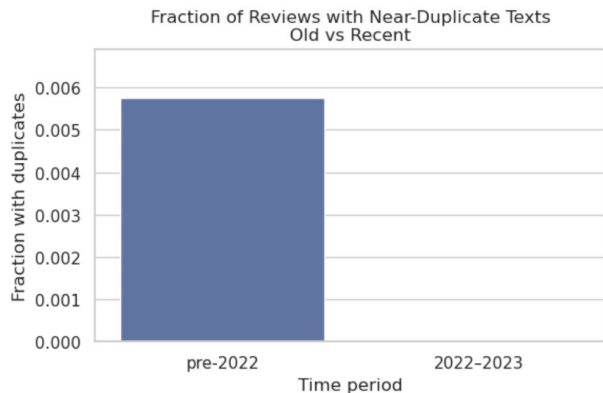
- Chart 1: Near-Duplicate Title Distribution
 - Majority of review titles only appear once or a few times, indicating mostly unique titles
 - Small number of titles appear to be repeated many times, indicating its presence is highly concentrated
- Chart 2: Fraction of Reviews with Duplicate Titles for Top 5 Books
 - Among the most reviewed books, 23-33% of reviews reuse near-duplicate titles
 - High-volume products are more likely to exhibit systematic title reuse

Near-Duplicate Review Texts (Books Category)



- Chart 1: Near-Duplicate Text Distribution
 - Near-duplicate review texts are extremely rare in the Books category (almost all of them have no duplicates)
- Chart 2: Fraction of Reviews with Duplicate Texts for Top 5 Books
 - Duplicate review text fractions remain below ~4% of total reviews
- **Conclusions: This suggests that while users may reuse short titles, they generally write original review bodies.**

Results: Effectiveness on Recent vs Old Reviews



- Similarity based methods detect more near-duplicate reviews in older data than in more recent reviews
- Fraction of reviews flagged as near-duplicates drops sharply with time
- Among reviews that do contain duplicates, older reviews show higher average duplicate counts
- **Conclusions:**
 - **Similarity based techniques are less effective at detecting duplication in recent reviews**
 - **Explanation: Recent reviews are more linguistically diverse, with more advanced and paraphrased AI content**

Conclusions:

- Amazon review activity has grown substantially over time, with clear seasonality and event-driven spikes
- A small subset of products and reviewers account for a disproportionate share of total reviews
- High review volume does not imply sustained engagement; many products experience brief peak periods
- Product price has little relationship with ratings, but lower-priced items receive far more reviews
- Review titles are frequently reused, especially for high-volume products, while review texts remain mostly unique
- Similarity-based methods are less effective on recent reviews, likely due to increased linguistic variation and advanced paraphrasing